

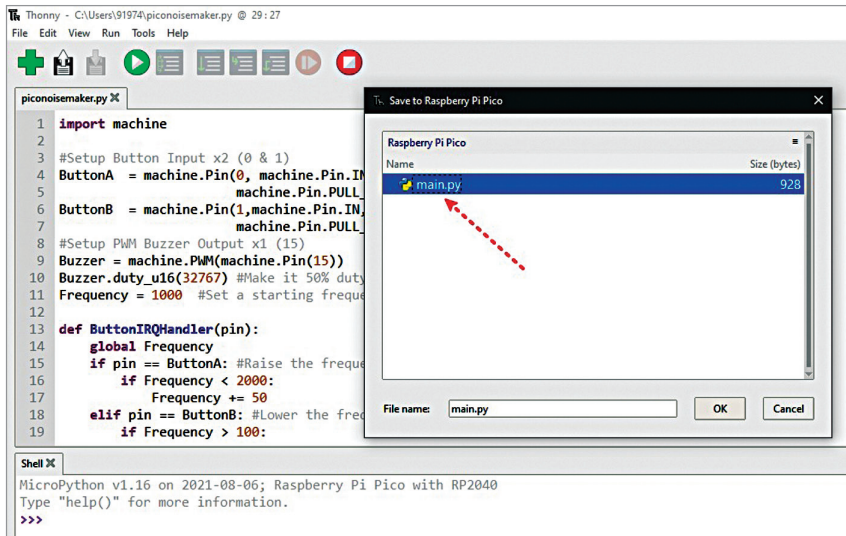


Fig. 6: Naming the Python code

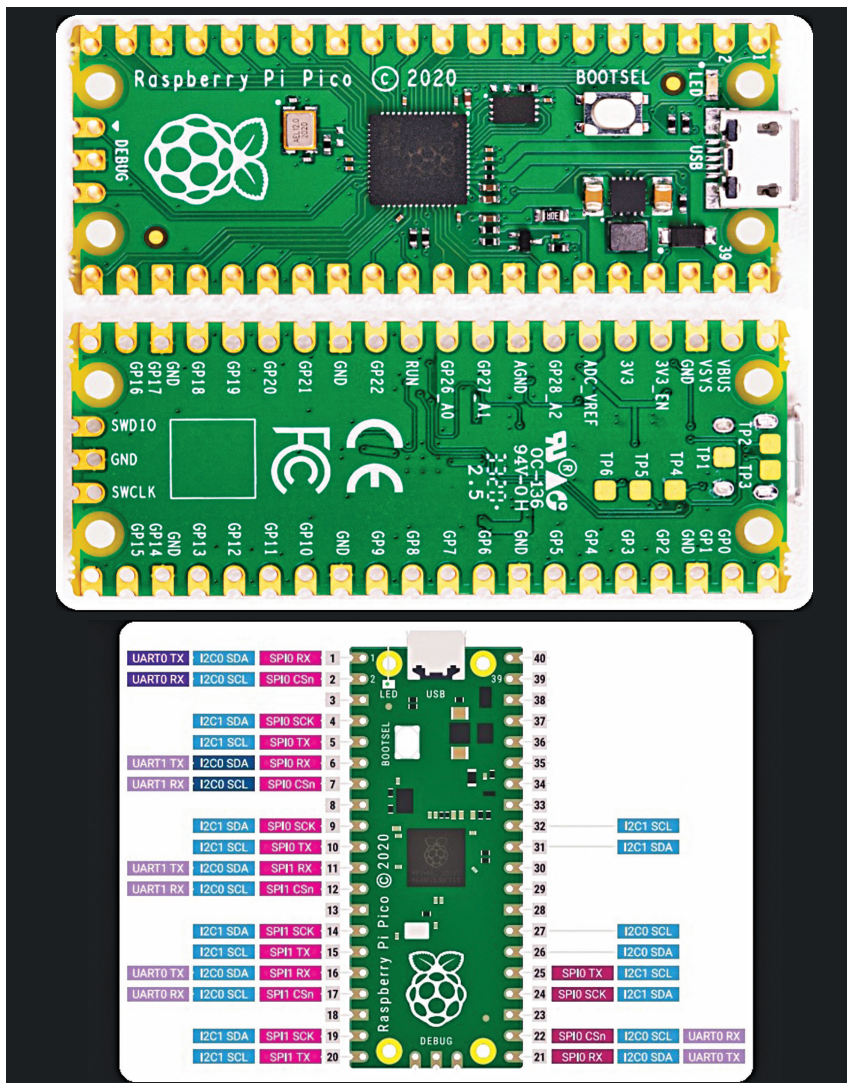


Fig. 7: Pin details of Raspberry Pi Pico

The simplistic code prepared for the above hardware setup to control the IRQ using two buttons (S1 and S2) and then generate an analogue output to drive the sounder element is given alongside (source of inspiration <https://kitronik.co.uk/>). The source code (`piconoisemaker.py`) is listed below:

```
import machine

#Setup Button Input x2 (0 & 1)
ButtonA = machine.Pin(0, machine.Pin.IN,
                      machine.Pin.PULL_DOWN)
ButtonB = machine.Pin(1, machine.Pin.IN,
                      machine.Pin.PULL_DOWN)

#Setup PWM Buzzer Output x1 (15)
Buzzer = machine.PWM(machine.Pin(15))
Buzzer.duty_u16(32767) #Make it 50% duty
                        cycle (32767/65535)
Frequency = 1000 #Set a starting
frequency of 1kHz

def ButtonIRQHandler(pin):
    global Frequency
    if pin == ButtonA: #Raise the frequency
        if Frequency < 2000:
            Frequency += 50
    elif pin == ButtonB: #Lower the
        frequency
        if Frequency > 100:
            Frequency -= 50

#Setup the IRQ and hook it to the handler
ButtonA.irq(trigger = machine.Pin.IRQ_
            RISING,
            handler = ButtonIRQHandler)
ButtonB.irq(trigger = machine.Pin.IRQ_
            RISING,
            handler = ButtonIRQHandler)

while True:
    Buzzer.freq(Frequency)
```

While on the subject, a PY file is a program file (or script) written in Python, which is an interpreted object-oriented programming language. It can be created and edited with a text editor but requires a Python interpreter to run. Simply go for Thonny!

Thonny IDE

Thonny comes with Python 3.7